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Álgebra Lineal

Grupo 19

Semestre 2022-2

$$10+1 = \textcircled{11}$$

Examen parcial no. 4

1)

$$P_1 = \{ax+b \mid a, b \in \mathbb{R}\}$$

$$K(ax+b) = 2bx + a - b$$

$$(\bar{p} \mid \bar{q}) = 2am + 2br$$

$$\begin{aligned} \bar{p} &= ax+b \\ \bar{q} &= mx+r \end{aligned} \in P_1$$

$$(\tau(\bar{u}) \mid \bar{v}) = (\bar{u} \mid \tau^*(\bar{v}))$$

$$(K(ax+b) \mid (mx+r)) = ((ax+b) \mid K^*(mx+r))$$

$$K(ax+b) = 2bx + a - b$$

$$K^*(mx+r) = \alpha rx + \beta m + \gamma r$$

$$\left[\underbrace{(2bx + a - b)}_{\substack{a' \\ b}} \mid \underbrace{(mx + r)}_m \right]$$

$$= \left[\underbrace{(ax + b)}_{\substack{a \\ b}} \mid \underbrace{(\alpha rx + \beta m + \gamma r)}_{\substack{m \\ r}} \right]$$

$$(\bar{p} \mid \bar{q}) = 2am + 2br$$

$$2(2bx)(m) + 2(a-b)(r) = 2(a)(\alpha r) + 2(b)(\beta m + \gamma r)$$

Nombre:

Día

Mes

Año

Folio

Tema:

$$(4b)(m) + (2a - 2b)(r) = (2a)(x) + (2b)(\beta m + \gamma r)$$

$$\underline{4bm} + \underline{2ar} - \underline{2br} = \underline{2ax} + \underline{2\beta bm} + \underline{2\gamma br}$$

$$2a = 2$$

$$\underline{a = 1}$$

$$2\beta = 4$$

$$\underline{\beta = 2}$$

$$2\gamma = -2$$

$$\underline{\gamma = -1}$$

$$K^*(mx + r) = ax + \beta m + \gamma r$$

$$\therefore K^*(mx + r) = rx + 2m - r$$

2)

$$T(x, y, z) = (2x + y, x + 2y, z)$$

$$\left. \begin{array}{l} \lambda_1 = 1 \\ \lambda_3 = 3 \end{array} \right\} \begin{array}{l} \text{Val.} \\ \text{Caro} \end{array}$$

$$E(1) = \{(a, -a, b) \mid a, b \in \mathbb{R}\} \Rightarrow P_1$$

$$E(3) = \{(c, c, 0) \mid c \in \mathbb{R}\} \Rightarrow P_2$$

$$x = a \quad y = b \quad z = c$$

Suma de los vectores por.

$$(a, -a, b) + (c, c, 0) = (x, y, z)$$

$$(a+c, -a+c, b+c) = (x, y, z)$$

$$a+c = x$$

$$a+c = x$$

$$-a+c = y$$

$$2c = x+y$$

$$c = \frac{x+y}{2}$$

Sust ② en ①

$$a+c = x$$

$$a = x - c$$

$$a = x - \left(\frac{x+y}{2} \right)$$

$$a = x - \frac{x}{2} - \frac{y}{2}$$

$$a = \frac{x-y}{2}$$

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Tema:

$$T = \lambda_1 P_1 + \lambda_2 P_2$$

$$T(x, y, z) = 1 \left(\frac{x-y}{2}, -\left(\frac{x-y}{2}\right), z \right)$$

$$+ 3 \left(\frac{x+y}{2}, \frac{x+y}{2}, 0 \right)$$

Descomposición
espectral

$$= \left(\frac{x}{2} - \frac{y}{2}, -\frac{x}{2} + \frac{y}{2}, z \right)$$

$$+ \left(\frac{3x}{2} + \frac{3y}{2}, \frac{3x}{2} + \frac{3y}{2}, 0 \right)$$

$$= \left(\frac{x}{2} + \frac{3x}{2} - \frac{y}{2} + \frac{3y}{2}, -\frac{x}{2} + \frac{3x}{2} + \frac{y}{2} + \frac{3y}{2}, z \right)$$

$$= \left(\frac{4x}{2} + \frac{2y}{2}, \frac{2x}{2} + \frac{4y}{2}, z \right)$$

$$T(x, y, z) = (2x + y, x + 2y, z)$$

Si
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